

AVAIS AHMED
ABQAIQ, DAMMAM

SUMMARY:

Senior Electrical Protection & Commissioning Engineer with 12+ years of extensive experience in HV/EHV substations, power transmission systems, protection relays, testing, commissioning, maintenance, and troubleshooting of 132kV, 220kV, and 380kV substations. Demonstrated expertise in protection schemes including Line Differential, Distance Protection, Transformer Differential, Busbar Protection, Feeder Protection, Breaker Failure Protection, Auto-Reclose, Synch Check, and Intertripping systems.

Possess strong hands-on experience with ABB, Siemens SIPROTEC, GE Multilin, Schneider Electric, and MICOM protection relays. Successfully led commissioning teams across utility, transmission, and industrial projects in Saudi Arabia and the Middle East while coordinating with clients, consultants, EPC contractors, and utility authorities. Extensive experience performing relay testing, fault investigations, disturbance analysis, root cause analysis, and system energization activities. CSD Approved professional with proven ability to deliver projects safely, efficiently, and within schedule.

WORKING EXPERIENCE:

Company: HUAWEI Technologies
Project: Solar Project, Saudi Arabia
Designation: Sr. ELECTRICAL ENGINEER
Period: JAN 2025 to Till Date

Duties & Responsibilities:

- Led testing and commissioning activities for multiple 380kV and 132kV substations involving line bays, transformer bays, reactor bays, and busbar extensions.
- Directed a team of protection engineers and technicians during energization of EHV substations while ensuring compliance with utility standards and project specifications.
- Performed end-to-end testing of ABB RED670 line differential relays including communication channel verification and protection logic validation.
- Commissioned Siemens SIPROTEC distance protection schemes for 380kV transmission lines including Zone-1, Zone-2, Zone-3 and teleprotection verification.
- Executed transformer differential protection testing on 380/132kV power transformers using ABB RET670 and GE Multilin relays.
- Conducted busbar protection scheme testing involving dynamic zone selection and CT supervision functions.
- Successfully resolved nuisance transformer differential trips caused by CT ratio mismatch through detailed relay configuration review.
- Investigated recurring feeder protection maloperations and identified wiring discrepancies within marshalling kiosks.
- Performed secondary injection testing using Omicron CMC356 and verified relay performance against approved setting files.
- Coordinated directly with utility representatives for CSD inspections and project approvals.
- Led SAT activities and prepared commissioning reports, punch lists, and energization packages.
- Troubleshoot IEC61850 GOOSE messaging failures impacting interlocking functionality between bays.
- Conducted fault record analysis following transmission line disturbances and verified relay operations using COMTRADE files.
- Improved commissioning efficiency by developing standardized testing procedures reducing testing time by 25%.
- Executed breaker failure protection testing ensuring proper trip initiation and backup isolation schemes.
- Validated synch-check functions prior to bus transfer operations.
- Coordinated outage planning and switching activities with system operation centers.
- Supported successful energization of multiple 380kV circuits without protection-related incidents.

8 EHV substations commissioned in 5 months

- led 15-person team through energization, CSD inspections, and SAT with **100% first-pass acceptance**.

DAILY IMPACT:

- Configured ABB RED670 line differential + fiber optic channel verification
- Commissioned Siemens SIPROTEC distance protection (Zone 1-2-3 + teleprotection)
- Tested RET670 + GE Multilin transformer differential schemes
- Validated busbar protection with dynamic zone selection
- Troubleshoot GOOSE failures → restored interlocking in 48 hours
- Performed COMTRADE analysis for fault investigations

SAVED:

- 120 man-days through standardized testing procedures
- \$2M+ by preventing transformer outages through CT ratio corrections
- 0 punch-list items during utility acceptance.

Company: National Transmission & Despatch Company Limited (NTDC)
Project: Dasu Hydropower Station, Upper Kohistan District, Khyber Pakhtunkhwa, Pakistan
Designation: Sr. ELECTRICAL ENGINEER
Period: JUL 2023 to DEC 2024

Duties & Responsibilities:

- Commissioned 132kV and 380kV transmission substations supporting national grid expansion initiatives.
- Performed configuration and testing of MICOM P443 line protection relays.
- Executed transformer protection testing including restricted earth fault and differential protection schemes.
- Verified breaker failure logic schemes and backup protection coordination.
- Conducted point-to-point wiring verification for protection panels and control systems.
- Investigated communication failures between protection relays and SCADA systems.
- Successfully restored protection communications affected by incorrect IEC61850 dataset mapping.
- Led relay replacement projects involving migration from legacy electromechanical systems.
- Performed fault analysis on line trips caused by CT saturation conditions.
- Supported factory acceptance testing activities with equipment manufacturers.
- Executed synchronization checks during transformer energization.
- Coordinated testing schedules with EPC contractors and client engineering teams.
- Verified auto-reclose schemes and intertripping logic across transmission corridors.
- Prepared detailed relay test reports and commissioning documentation.
- Reduced commissioning punch-list items through proactive quality inspections.
- Conducted relay setting validation based on approved coordination studies.
- Assisted with disturbance recorder configuration and event retrieval.
- Supported utility witness testing and final project handover activities.

Company: Power China
Project: Dasu–Mansehra–Islamabad Transmission Line Project — 765/220 kV Mansehra Grid Station, Pakistan
Designation: ELECTRICAL SITE ENGINEER
Period: MAY 2020 to JUNE 2023

Duties & Responsibilities:

- Led commissioning teams across multiple HV substation projects.
- Conducted comprehensive testing of ABB and Siemens protection relays.
- Commissioned transformer differential protection for large power transformers.
- Performed line differential scheme testing using fiber optic communication channels.
- Troubleshoot relay non-operation issues caused by incorrect binary input assignments.
- Investigated transformer trips caused by CT polarity reversal and implemented corrective actions.
- Conducted dynamic busbar protection testing.
- Executed end-to-end relay testing for transmission line protection schemes.
- Verified teleprotection channels and communication redundancy systems.
- Supported protection coordination studies and setting calculations.
- Performed relay firmware upgrades while maintaining system reliability.
- Led outage execution activities for protection modifications.
- Validated breaker failure operation times against utility requirements.
- Assisted engineering teams with protection design reviews.
- Conducted root cause analysis following protection system alarms.
- Supported SAT and client witness testing activities.
- Implemented testing quality procedures improving first-pass acceptance rates.
- Delivered technical training to junior commissioning engineers.

Company: SUNIR
Project: Construction of 500 kV & 220 kV Transmission Lines and 500 kV Grid Station at Shikarpur Project, Pakistan
Designation: ELECTRICAL ENGINEER
Period: JUN 2018 to APR 2020

Duties & Responsibilities:

- Performed testing and commissioning of 132kV substations.
- Configured Siemens SIPROTEC protection relays for feeder and transformer applications.
- Conducted secondary injection testing using Omicron equipment.
- Troubleshoot feeder protection trips caused by incorrect CT burden calculations.
- Executed busbar protection scheme verification.

- Supported commissioning of substation automation systems.
- Performed cable continuity and insulation testing activities.
- Validated protection logic diagrams and control schematics.
- Coordinated with contractors during construction and commissioning phases.
- Conducted relay settings implementation and verification.
- Assisted in fault investigations and event analysis.
- Supported SCADA integration testing and point verification.
- Executed breaker control logic testing.
- Participated in transformer energization procedures.
- Performed acceptance testing for new protection panels.
- Developed commissioning records and testing documentation.
- Improved system reliability through proactive relay maintenance programs.
- Supported client technical review meetings.

Company: Imperial Construction Company(ICC) Pvt Ltd:

Project: Procurement of Plant: Design, Supply, Installation, Testing and Commissioning of 220kV Double Circuit Twin Bundle Uch - Sibbi Transmission Line – Turnkey, Pakistan

Designation: ELECTRICAL ENGINEER

Period: JAN 2016 to MAY 2018

Duties & Responsibilities:

- Supported protection commissioning activities for HV substations.
- Tested feeder protection and overcurrent schemes.
- Conducted transformer differential relay testing.
- Performed point-to-point verification activities.
- Assisted senior engineers during energization activities.
- Troubleshoot relay alarm conditions and communication failures.
- Conducted CT and VT polarity verification.
- Executed cable megger and continuity testing.
- Supported fault record retrieval and event analysis.
- Verified breaker failure protection logic.
- Conducted SCADA point validation tests.
- Prepared testing documentation and reports.
- Assisted with relay setting implementation.
- Participated in FAT activities.
- Supported utility inspections and approvals.
- Maintained testing equipment calibration records.
- Conducted preventive maintenance activities.
- Assisted with relay retrofit projects.

Company: MEP SOLUTIONS

Project: Rehabilitation of the 22MW Jabban Hydroelectric Power Station, Pakistan

Designation: MEP ELECTRICAL ENGINEER

Period: SEP 2013 to DEC 2015

Duties & Responsibilities:

- Supported electrical testing and commissioning activities.
- Conducted relay secondary injection tests.
- Assisted with substation protection system verification.
- Performed electrical panel inspections.
- Executed cable testing and termination checks.
- Supported transformer commissioning activities.
- Conducted equipment functional testing.
- Verified protection wiring against approved drawings.
- Assisted with fault investigations.
- Prepared daily commissioning reports.
- Supported relay configuration activities.
- Conducted breaker operational tests.
- Participated in site acceptance testing.
- Assisted with client inspections.
- Maintained testing documentation and records.
- Supported preventive maintenance programs.
- Performed protection panel troubleshooting.
- Assisted senior engineers during project execution.

CORE COMPETENCIES:

- HV/EHV Substations (132kV / 220kV / 380kV)
- Protection Relay Testing & Commissioning
- Distance Protection Schemes
- Transformer Differential Protection
- Busbar Protection Systems
- Feeder Protection Schemes
- Line Current Differential Protection
- Breaker Failure Protection
- Auto-Reclosing Systems
- Synch Check Schemes
- Protection Coordination Studies
- Fault Analysis & Disturbance Investigation
- Substation Automation Systems
- IEC 61850 Engineering
- SCADA Integration
- Relay Configuration & Settings
- FAT & SAT Execution
- Site Team Leadership
- Client Coordination
- Root Cause Analysis
- Electrical Troubleshooting
- Power System Reliability Improvement
- CSD Compliance Requirements

SKILLS:**Protection Relays**

- ABB RED670, REL670, RET670, REC670, REF615
- Siemens SIPROTEC 4 & SIPROTEC 5 Series
- GE Multilin UR Series
- Schneider Easergy P3/P5
- MICOM P443, P444, P546, P541
- SEL Protection Relays

Testing Equipment

- Omicron CMC356
- Omicron CMC256
- Omicron Test Universe
- Freja Relay Test Sets
- Megger Equipment
- Secondary Injection Test Kits

Software

- DIGSI
- PCM600
- CAP505
- MiCOM S1 Studio
- EnerVista
- ETAP
- AutoCAD
- IEC 61850 Tools

CERTIFICATIONS

- CSD Approved Engineer
- IEC 61850 Substation Automation Certification
- Omicron Protection Relay Testing Certification
- High Voltage Safety Certification
- Transmission Substation Commissioning Certification

MAJOR PROJECT HIGHLIGHTS

- 380kV GIS Substation Commissioning – Saudi Arabia
- 132kV Transmission Line Protection Upgrade Project
- 380/132kV Power Transformer Protection Commissioning
- National Grid Expansion Protection Systems Project
- IEC61850 Digital Substation Integration Project

- EHV Busbar Protection Modernization Program

PROFESSIONAL ACHIEVEMENTS

- Successfully commissioned more than 50 HV/EHV substations.
- Tested and commissioned over 2,000 protection relays.
- Supported energization of 132kV and 380kV transmission networks without major protection failures.
- Reduced commissioning punch-list items by 30% through enhanced quality controls.
- Led multidisciplinary teams across major utility transmission projects.
- Recognized by clients for excellence in protection troubleshooting and commissioning execution.

Additional Information:

- Excellent English proficiency.
- Ability to work under pressure and meet deadlines.
- Open to relocation for international projects.